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# Design and development of a wrist rehabilitation device with an interactive game

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ABSTRACT

In this paper, a wrist rehabilitation device integrated with an interactive game is developed. The device is designed for flexion/extension and ulnar/radial motions. The motion of the wrist joint is measured by an optical encoder. A hall sensor is also used to measure the force emitted from the DC motor. These data are fed back to the interactive gaming environment to create the movement of goals in the video game. There are two types of interactive games designed. The first requires the practitioner to move their wrist to achieve a specific goal. The second exercise requires the practitioner to move their wrist joint continuously to reach moving targets. In the game interface, therapists can easily adjust training intensity, training time, and change training limits depending on the recovery status of the patients. At the same time, it also records the patient's training efforts and monitors visual graphs of performance during exercise. The experiments were conducted to tune the games and to get a preliminary evaluation of the device. The feedback from the patients and therapists are recorded to improve the device and game interfaces. Game parameters are adjusted during the patient's exercise process under the supervision of a physiotherapist. The values of rotation angle and torque are collected and then presented in graphs so that physiotherapists can assess the patient's recovery status. The results show that the patients enjoy the games. The therapist is satisfied with the game interface and says that the games are intuitive and easy to play.

## 1. Introduction

Every year, around the world, more than 15 million people are at risk of having a stroke. The effects of a stroke can be short-term or long-term, depending on which part of the brain is affected and how quickly it is treated. Stroke survivors can experience widespread disabilities including difficulties with mobility and speech, as well as how they think and feel. Stroke is one of the main factors leading to impaired motor function of the human upper limb. These patients are significantly limited in their daily social and family activities. Stroke patients require appropriate post-injury care and rehabilitation therapy to restore lost abilities and help them return to normal daily activities [1–3]. Typically, this is achieved through a long-term regimen of intensive and repetitive rehabilitation therapy. The rehabilitation therapy requires supervision, monitoring and support from physiotherapists. Another problem is that conventional recovery methods lack quantitative feedback on performance. Therefore, it is difficult to evaluate upper limb mobility without corresponding supporting data. This factor stimulates the application of alternative rehabilitation methods [4–7].

The idea of using robots in rehabilitation therapy has helped solve the above problems. Therefore, many rehabilitation robot systems have been developed [8–12]. The robots are designed to create precise, repetitive movements and can be adjusted to suit the patient's motor abilities. Rehabilitation robots help reduce physical therapists' efforts, and they can even provide longer and more intense therapy sessions. Based on integrated sensors, the patient's recovery status is also more accurately assessed. Thanks to that, the robot provides exercises with intensity appropriate to the recovery condition [13,14]. Rehabilitation robots can be also equipped with a virtual environment to provide an entertaining and motivating background for patients, which can lead to an increment of exercise intensity. For each different damaged limb, there will be robots designed to suit the activities of each limb. Some

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rehabilitation robots have been developed for various limbs such as the shoulder, arm, wrist, hand, finger, knee, and ankle. The hand plays an important role in performing activities in daily life. The wrist joint connects the hand to the forearm, helps position complex hand poses, and creates complex and flexible movements in daily activities. Therefore, when the wrist joint is damaged, tasks that require flexibility and dexterity are difficult to perform [15–17]. Wrist joint rehabilitation helps patients restore the movements of their wrist joints for daily activities and work. It is as important as hand rehabilitation.

The development of wrist rehabilitation robots is a combination of mechanical design, actuators, sensors, control algorithms, software, and exercise design. CR2-Haptic [18] is a portable and compact device for wrist joint rehabilitation. This design used a DC brushed motor to create the rotation motion for the handle. The rotation angle and motor current were measured by a rotary encoder and current sensor, respectively. Various rehabilitation strategies have been designed in a virtual reality environment to enhance the patient recovery process. In Ref. [19], a Universal Haptic Drive device was designed to create a 2-DOF movement for either arm or wrist joint. The force of two sets of DC motors is controlled by an impedance control algorithm in the training movement. Lee et al. [20] developed a robot wrist that was actuated coaxial spherical parallel mechanism. This design has advantages such as low moving parts inertia, high torque output, and high dynamic motion response. Parallel mechanisms for wrist and forearm rehabilitation are also proposed in RiceWrist [21] and CRAMER [22] robots. Martinez et al. [23] developed a Wrist Gimbal device with 3-DOF for the forearm and wrist rehabilitation. The device used the Maxon DC servo motors to create motion in position trajectory tracking tasks of passive exercises and generate resistive torques in active exercises. The control system used a Quanser Q8-USB data acquisition board together with Matlab/-Simulink and Quanser QuaRC software to read data from sensors. Another design with 3-DOF motion was also proposed in Ref. [24] called Physiotherabot/WF. This robot used the Hybrid Impedance Control (HIC) method in the control system, and it can perform passive, active assisted, isotonic, isometric and isokinetic exercises. NU-Wrist robot is a novel self-aligning 3-DOF robotic system for wrist and forearm rehabilitation that was proposed in Ref. [25]. The robot was designed with a novel compliant robot handle that can self-align the axes wrist joints and rotation axes of the robot during therapy exercises. Dynamixel MX-106 and MX-28 servomotors were used as joint actuators in this prototype to simplify electronic circuits and control algorithms. Lue et al. [26] developed a compact wrist rehabilitation robot that uses a curvilinear guide. A "moving window" control strategy based on impedance control is proposed and implemented on the robot.

In recent years, interactive games have been rapidly applied as new treatment approaches in stroke rehabilitation. More and more game consoles are being applied in clinical treatment. Interactive games are designed to help patients reacquaint themselves with real-life movements and situations in everyday life. Key requirements for motor learning include intensity, frequency, repetition, and task-oriented training can be provided by the software, the interaction between the patient and the game environment makes the patient feel involved in their rehabilitation. Thanks to the adjustment of the degree of difficulty, as well as options in gaming tasks, the virtual systems can accommodate patient needs, abilities, and goals. Video games makes recovery sessions more interesting and engaging for goal-oriented tasks. As the stroke patient's functional ability progresses, tasks can be easily upgraded using manipulations in a safe environment for the patient [27,28]. Furthermore, the games can be performed without the supervision or intervention of physical therapists. Patients do not need to travel to physical therapy centers to perform exercise sessions. Instead, the therapy process can be performed regularly at home [29]. A recent randomized controlled trial suggests that using commercial video game therapy improves balance, postural control, functionality, quality of life, and motivation in patients with subacute stroke [30,31].

In this paper, a wrist rehabilitation device integrated with an interactive game is developed. The device is designed for flexion/extension and ulnar/radial motions. The motion of the wrist joint is measured by an optical encoder. This information is fed back to the interactive gaming environment to create the movement of goals in the video game. Thus, creating interaction between the patient's movement and the virtual environment. Additionally, a hall sensor is also used to measure the force emitted from the motor. Depending on the patient's condition, the force value will be controlled at an appropriate level. The game interface will be designed so that users can easily adjust training intensity, training time, change training limits depending on recovery status. It also provides visual graphs of performance during exercise, from which physical therapists can accurately assess the patient's condition and recovery level.

## 2. Prototype design and fabrication

Fig. 1 shows the design of the prototype for the wrist rehabilitation device. The design is quite simple. It consists of a DC motor to drive a crank. The crank is connected directly to the shaft of the DC motor through a flange coupling. To measure the rotation angle of the motor, an encoder is used. The motion of the motor is transmitted through the encoder through the cable-driven. A rotation and slider joint between the crank and the handle help change the direction of the wrist motion. In Fig. 2, the handle is vertical, providing flexion/extension motion. In Fig. 3, the handle is horizontal, providing ulnar/radial motion. After adjusting the appropriate position for the handle, it is fixed to the crank by tightening the 2 bolts on the crank to secure the handle. The human forearm will be placed on a pedestal and held in place by an adjustable fabric strap.

Fig. 4 shows the realistic prototype. The components for mounting the motor and the crank are made from aluminium alloy to ensure rigidity. The handle is made using 3D printing technology with a steel shaft at the end. The pedestal is also made with a 3D printer.

The motor used is the XD340 motor, it can provide a maximum torque of 4Nm. It operates at a voltage of 12 V, and the maximum current is 3 A. The encoder used to measure the rotation angle of the motor is E6B2-CWZ6C 4000P/R 0.5 M. The resolution of the encoder is 400 pulse/revolution.

In this project, to control the motor, the Arduino Uno board is used. The function of the Arduino board is that read signals from sensors to control the movement of the motor. In this project, a current sensor is used to determine the torque of the motor. Therefore, we can adjust the output torque of the motor. Fig. 5 shows the wiring circuit of the electronic system.

The current sensor used is the Hall sensor ACS712 5 A. The current sensor is connected in series with the motor driver. The output of the current sensor is an analog signal. So, the output pin of ACS712 is connected to the analog pin A0 of Arduino. The output value from the

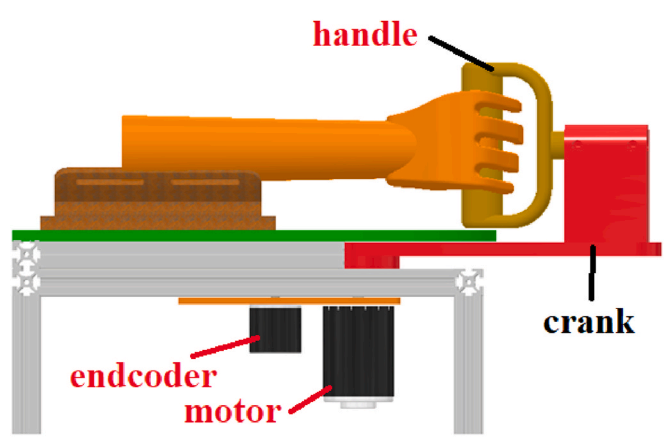


**Fig. 1.** The prototype for the wrist rehabilitation device.

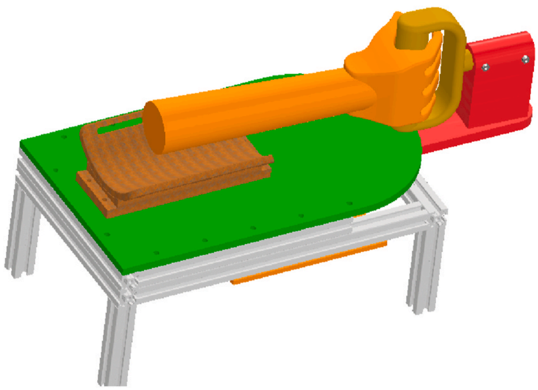

**Fig. 2.** Change the direction of the handle to vertical for flexion/extension motion.

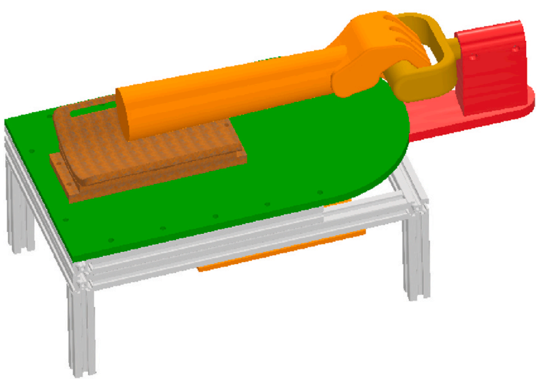

**Fig. 3.** Change the direction of the handle to vertical for redial/ulnar motion.

**Fig. 4.** The realistic prototype.

sensor is the voltage that is linear to the current to be measured with a sensitivity of 63–190 mV/A. To control the motor, the BTS7960 driver is used.

To create impedance, the motor will be torque controlled (control the current) using a PID controller:

$$PWM = K_p e(t) + K_d \frac{de(t)}{dt} + K_i \int e(t) \quad (1)$$

where $e(t)$ is the error between the measured torque $\tau_m(t)$ and the desired torque $\tau_d(t)$:

$$e(t) = \tau_m(t) - \tau_d(t) \quad (2)$$

Since the torque of a DC motor is proportional to the current ($\tau = ki$), it is possible to determine the value of the motor's torque by a current sensor. The torque of the motor is not controlled at a fixed value but changes according to the position of the motor. The desired torque $\tau_d(t)$ will depend on the rotation angle and angular velocity of the motor:

$$\tau_d(t) = k\alpha(t) + b\omega(t) \quad (3)$$

where k is the stiffness coefficient, b is the damping coefficient, $\alpha$ is the rotation angle of the motor measured from the encoder, $\omega(t)$ is the angular velocity of the motor. The angular velocity of the motor is not directly measured but will be determined from the rotation angle of the motor:

$$\omega(t) = \frac{x(t+\Delta t) - x(t)}{\Delta t} \quad (4)$$

It can be seen that when the motor rotates at a larger angle, the larger torque generated will pull the motor back to its original position. The analog signal from the current sensor must be filtered for noise before being used in the controller. The averaging filter and the compensation filter are used in combination to reduce noise from the sensor. The Arduino will first read 50 consecutive analog values from the sensor and take the average:

$$ADC = \frac{\sum_{i=0}^{50} ADC(i)}{50} \quad (5)$$

Then, the compensation filter is applied:

$$ADC = \alpha.ADC + (1-\alpha)Last\_ADC \quad (6)$$

The obtained ADC value will be converted to the current value. The analog-to-digital converter (ADC module) in the Arduino has a resolution of 10 bit, so the ADC value will be from 0 to 1023. The ACS712 sensor will output a voltage from 0 V to 2.5VDC, corresponding to a current from 0 A to 5 A, we can determine the formula to convert from ADC value to current as follows:

$$I = 5\frac{(512 - ADC)}{512} \quad (7)$$

Fig. 6 shows the step-by-step implementation of the motor control algorithm. First, the Arduino will read the motor rotation angle from the encoder, this angle value will be used to calculate the desired torque value $\tau_d(t)$ according to equations (4) and (3). The Arduino then reads the motor current value from the current sensor. The current value is converted to the motor torque to calculate the error according to equation (2). The PID controller according to equation (1) will be applied to calculate the PWM value to be output to control the motor. Finally, the Arduino will output the PWM signal to the driver to control the motor.

The values of the rotation angle and torque of the motor will be sent to the computer for display on the exercise interface to aid in the rehabilitation process. Data communicates between the computer and the Arduino via asynchronous serial communication (UART) with a set baud rate of 38400 bits/sec.

## 3. Development of game interfaces for rehabilitation

The game interface is designed using Python programming language with the support of the Tkinter library. Tkinter is a Python library that

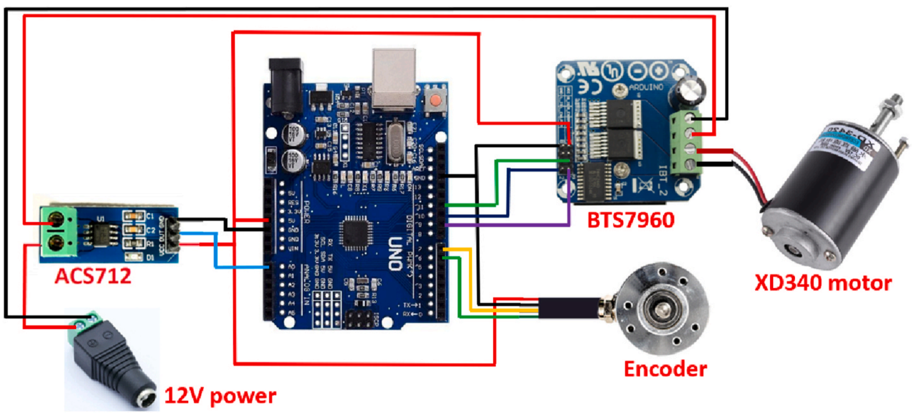


**Fig. 5.** Wiring circuit to connect electronic devices to each other.

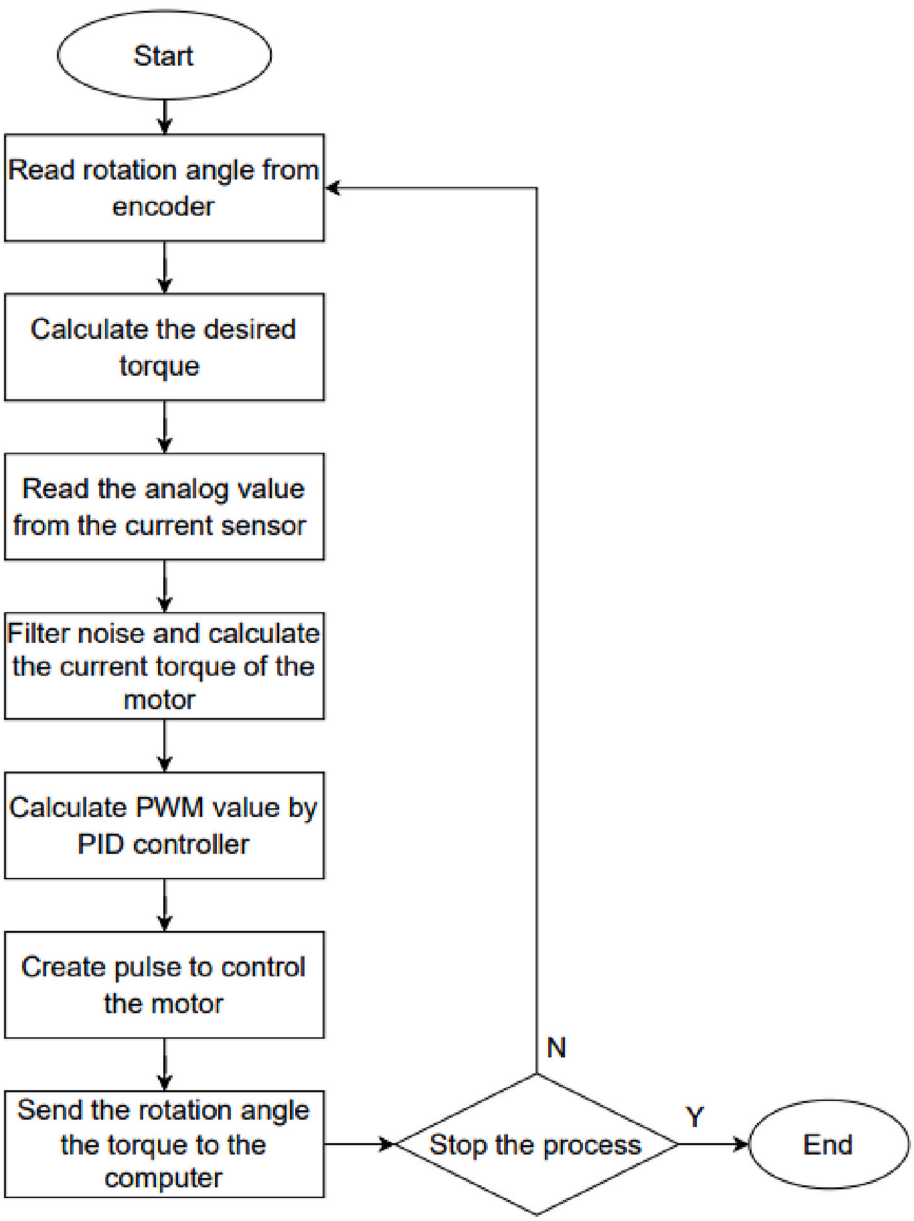


**Fig. 6.** The flowchart of the control algorithm.

can be used to construct a basic graphical user interface (GUI) applications. To communicate between Arduino and the Python program, the pySerial module is used to access and work with serial ports.

Fig. 7 shows the main interface screen of the game designed using Python programming. This screen allows the user to select exercises and select different levels of exercise. There are 6 different levels of practice. In which level 0 will not be the resistance from the motor. At other levels, the resistance created by the motor will change as the level increases. The resistance is changed by varying the coefficients k and b in Equation (3). When the user chooses to change the level, the computer will send down the characters corresponding to the selected level. The Arduino receives the character and will change the k and b coefficients according to the received character.

Exercises are divided into two categories. The first requires the practitioner to move the wrist to achieve a specific goal. When the practitioner rotates the wrist joint, the cat on the screen will move accordingly. The trainer's task is to make the cat achieve the goals in turn, when the final goal is reached, it will return to the original position. Fig. 8 illustrates this exercise. The cat is moved to eat three small fishes and a big fish. Practitioners can choose the limit of the rotation angle to suit their condition. This interface screen has three buttons. The "start" button is used to initiate an exercise when the practitioner is ready. When the practitioner wants to increase the resistance of the motor, he can press the "next level" button. The "back" button is used to return to the main screen. This exercise is suitable for patients in the early stages of recovery. The patient will try to move the wrist to achieve the greatest possible rotation (without resistance from the motor). After the patient has flexed and extended the wrist to the maximum angle, it is possible to switch to exercise with increasing motor resistance to improve wrist strength.

The second exercise requires the practitioner to move the wrist joint continuously to reach moving targets. Objects will change position continuously on the screen. Fig. 9 shows the screen interface of this exercise. The player's task is to move the basket to catch red apples falling in different random positions on the screen. After each time a red apple is caught, the score will increase by one unit. The maximum score will be set in advance. Players complete the mission when they reach the set score. In addition to red apples, there will also be green apples appearing on the screen. If the player catches a green apple, two points will be deducted. After a specified time, if the player does not achieve the required score, it will be considered as not completing the mission.

## 4. Experiment results

The experiments were performed to tune the game and to get a preliminary evaluation of the device. The experimental process is performed on patients under the supervision of a physiotherapist. Three patients, two women and one man, participated in the rehabilitation

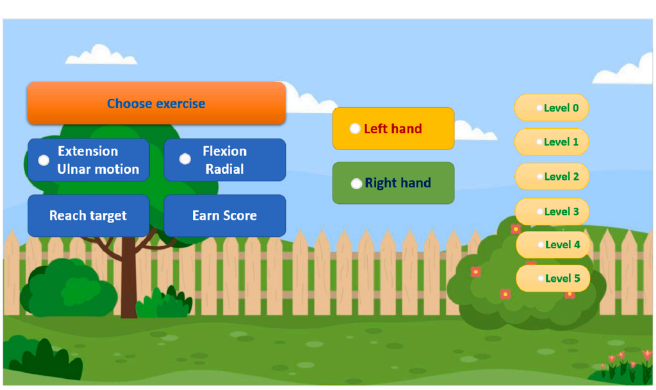


**Fig. 7.** The main screen of the game was designed using Python programming language.


**Fig. 8.** The interface screen of the reach target exercise consists of three small targets and a big final target.

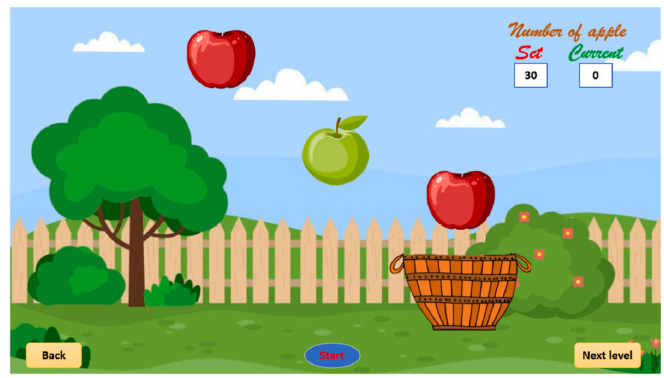


**Fig. 9.** The interface screen of the earn score exercise.

process with the device in three days. Initial feedback from the patients and therapist helped improve the device and the game interface. For example, in the first test, the device could not be used on female patients recovering from a stroke. The reason is that the first design of the device used a motor with a gearbox. The patient has just been injured and her wrist strength is still very weak so she cannot rotate the motor (due to the resistance and friction of the gears in the gearbox). Therefore, we replaced it with another motor without a gearbox. The original game was also designed with a simple, unattractive interface that did not create excitement for patients. After consulting with the physiotherapist, we redesigned the game, making it more lively and attractive. The physiotherapist also requested that the game should be divided into different levels to suit the recovery status of the patients.

The games are designed to ensure operability. However, to evaluate the device's suitability for the rehabilitation of patients, experiments need to be performed to adjust the parameters in the game. With the designed games, the parameters that need to be adjusted are the limit of the wrist's angle, the maximum resistance torque of the device, the goal of each level, and the range of change in the position of targets.

First, the limited parameters of a few healthy adults were recorded. The maximum flexion/extension rotation angle is $80^\circ$, redial/ulnar rotation angle is $20^\circ$. The maximum torque for flexion/extension movement is 4 N.m. These parameters are used for the highest level of the game (level 5). The game is divided into 6 levels. The levels are not evenly spaced but must be divided into levels appropriate to the patient's condition. According to advice from a physiotherapist, for patients who have recently suffered an injury, the recovery process will be slow. Therefore, we set the limit of the levels 0, 1 and 2 to be close to each other. The testing process on three patients helped determine the limits of each level. Table 1 shows the value of each level.

In the early stage, the patients can rotate the wrist to reach an angle

**Table 1**
The limit of rotation angle and resistance torque.

| | Wrist angle | Resistance Torque |
|---|---|---|
| **Level 0** | 0–50° | 0 Nm |
| **Level 1** | 0–50° | 0.5 Nm |
| **Level 2** | 0–50° | 1 Nm |
| **Level 3** | 0–70° | 2 Nm |
| **Level 4** | 0–70° | 3 Nm |
| **Level 5** | 0–80° | 4 Nm |

up to 50°, however, the wrist strength is still weak. The exercise process will help patients gradually increase wrist strength as well as the ability to control movement. The goal of rotation angle will not be fixed, depending on the patient's ability, the physiotherapist will enter the maximum rotation angle value into the game.

After adjusting the game parameters, experiments are performed to obtain preliminary assessments showing the device's suitability during the patient's recovery. Experiments on both normal people and patients have been performed. The results are shown in Figs. 10–12. The graphs show the rotation angle of the wrist joint and the motor torque over time during an exercise to achieve the required goal.

Fig. 10 shows the results of a normal person. For undamaged wrist joints, the movements are flexible, and the wrist torque is strength, so the target angle can be quickly achieved. The practitioner only needs about 1.5s to rotate the wrist to reach the required angle. The graph in Fig. 10 will be used as a reference to evaluate the practitioner during the rehabilitation period.

Fig. 11 is the result of a patient for the first time using the device. Based on the graph, it can be seen that it is quite difficult for the patient to move the wrist joint to the required angle value. The patient needs to stop at intermediate goals before reaching the final goal, so the graph has a ladder shape. The time to complete the request is therefore also extended up to 6s. The patient also cannot maintain the maximum rotation for a long enough time. Due to the pulling force from the motor, the patient's wrist angle decreased from 35° to 30° in the maintenance time. Comparing the graph between a normal person and a recovering person, we can evaluate the patient's condition. In the early stages, the patients will not be able to control the movement of the wrist joint, so the movements will be interrupted. The patients will move in small steps to reach the target angle. After a period of practice, the patient will gradually control the movement and strength of the wrist. Movements become smoother and graphs also become smoother. Fig. 12 shows results after a period of practice with the device of the above patient. The graph shows that the patient has gradually become familiar with the device. Even though it was only for a short time, it also showed improvement in wrist joint mobility. The time to reach the required rotation angle has been reduced to 4s for two days of exercise. Although there are still interruption motions, the stopping time at intermediate points is also reduced. If compared with the graph of the hand joint of a person who has not suffered damage, this patient's condition has not yet reached a normal state. At this stage, the patient can switch to continuous exercise mode to continue improving the flexibility and movement speed of the wrist joint.

Continuous exercises will help patients improve wrist flexibility. Patients should start with a low score and gradually increase the score during practice to gradually adapt. The movement speed of the targets in this game is also adjusted to match the movement speed of the patient's wrist joint. The speed will gradually increase with each level.

Experiments have shown that patients enjoy participating in rehabilitation exercises with the device. This makes rehabilitation exercises entertaining and engaging and keeps the patients motivated longer. The therapists were satisfied with our game interface and she said that the games were intuitive and easy to play so patients did not need much training.

## 5. Conclusion

This paper has developed a wrist robot for rehabilitation therapy. The device is designed to easily switch between flexion/extension motion and ulnar/radial motion, and the handle position is also adjustable to fit a variety of patients. The device is used by patients for active exercises. The motor will create resistance force against the patient's movement during therapy. The resistance force increases according to the wrist joint angle. The current sensor and encoder are used in the electronics circuit to measure the motor current and rotation angle, respectively. The desired torque is calculated from the position of the wrist. The PID controller is applied to the motor can create the torque equal to the desired value.

A game interface has been designed to create an interactive environment and make the patient feel involved in their rehabilitation. The gaming interface has 6 levels to choose from, the resistance created by the motor increases with each level. So, the patient can easily adjust training intensity, training time, change training limits depending on recovery status. The software also records the patient's training efforts and monitors visual graphs of performance during exercise, from which physical therapists can accurately assess the patient's condition and recovery level. Experimental results were performed to tune the game and get a preliminary evaluation of the device. Initial feedback from the patients and therapist helped improve the device and the game interface. To make the device can be used in rehabilitation process, some parameters of the games were adjusted such as the parameters that need to be adjusted are the limit of the wrist's angle, the maximum resistance torque of the device, the goal of each level, and the range of change in the position of targets.

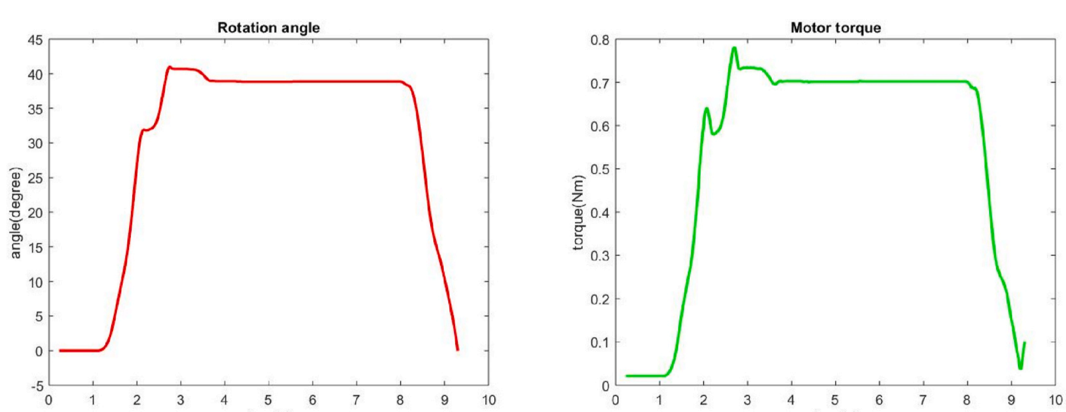


**Fig. 10.** Result of a normal person using the device to conduct extension motion to reach the target at the position of 40°. The left picture is the rotation angle and the right is the resistance torque of the motor.

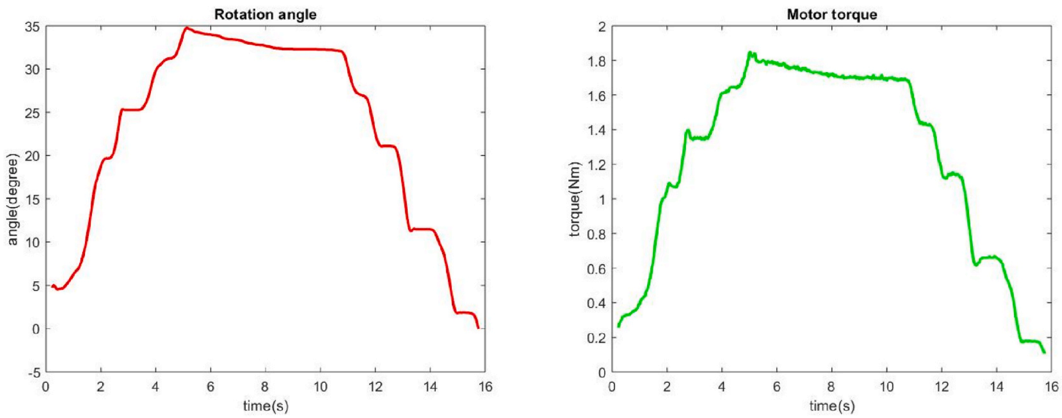


**Fig. 11.** Result of a patient in the first time using the device to conduct the extension motion to reach the target at the position of 35°.

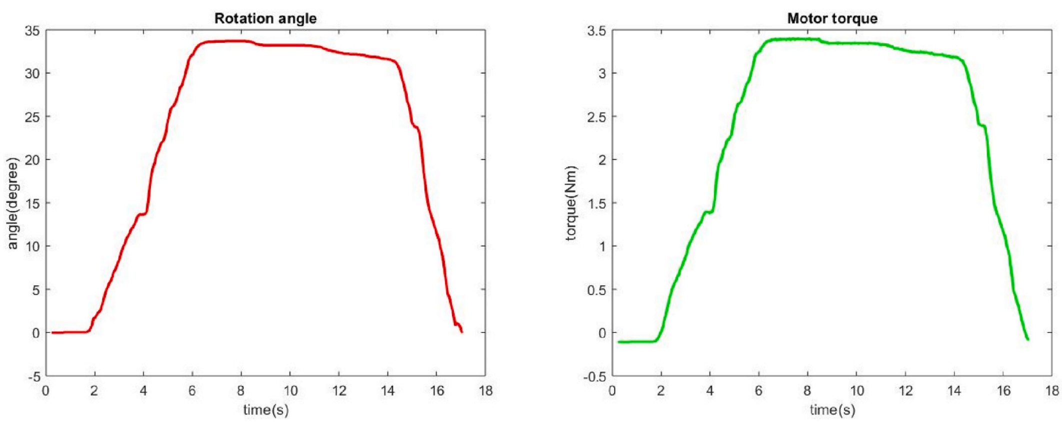


**Fig. 12.** Result of a patient after using the device for a while.

After adjusting the game parameters, experiments are performed to obtain preliminary assessments showing the device's suitability during the patient's recovery. Experiments on both normal people and patients have been performed. The normal person only needs about 1.5s to rotate the wrist to reach the required angle and the graph of motion during exercise is very smooth. Patients who exercise for the first time find it more difficult to achieve the goal; it takes them more than 6s to complete the task. They cannot move the wrist joint continuously and need to pause midway to gain momentum. After a period of practice, the patient will gradually control the movement and strength of the wrist. The results show that the time to reach the required rotation angle has been reduced to 4s for two days of exercise.

## CRediT authorship contribution statement

**Cong Vo Duy:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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